

ContactManager.java

```
1 import java.util.*;
2
3 public class ContactManager {
4
5     public static void main(String[] args) {
6
7         HashMap<String, Contact> contacts = new HashMap<>();
8
9         contacts.put("Ada Lovelace", new Contact("Ada Lovelace", "+1 617 555 0101"));
10        contacts.put("Grace Hopper", new Contact("Grace Hopper", "+1 125 555 1234"));
11        contacts.put("Alan Turing", new Contact("Alan Turing", "+1 234 567 4321"));
12        contacts.put("Linus Torvalds", new Contact("Linus Torvalds", "+1 999 555
4444"));
13        contacts.put("Margaret Hamilton", new Contact("Margaret Hamilton", "+1 111 222
3344"));
14
15        Contact foundContact = contacts.get("Ada Lovelace");
16        if (foundContact == null) {
17            System.out.println("Contact not found.");
18        } else {
19            System.out.println(foundContact.getName() + ": " +
foundContact.getPhone());
20        }
21
22        Contact missingContact = contacts.get("Unknown Person");
23        if (missingContact == null) {
24            System.out.println("Contact not found.");
25        } else {
26            System.out.println(missingContact.getName() + ": " +
missingContact.getPhone());
27        }
28
29        ArrayList<Contact> sorted = new ArrayList<>(contacts.values());
30        sorted.sort((a, b) -> a.getName().compareTo(b.getName()));
31
32        System.out.println("=== All Contacts ===");
33        for (Contact contact : sorted) {
34            System.out.println(contact.getName() + ": " + contact.getPhone());
35        }
36    }
37 }
```